CAGE-Fake: Article Trustworthiness Audit Report

Calibration-Aware Gating and Evidence-Grounded Reasoning for Trustworthy Fake News Triage

LEGAL & METHODOLOGICAL DISCLAIMER: *A trained fake-news classifier predicts linguistic, stylometric, and semantic patterns learned from benchmark datasets; it does **NOT** independently verify whether a news story is factually true. URL extraction, model pattern recognition, and external fact verification remain strictly separate processes. High uncertainty cases are flagged for professional human editorial review.*

Source / URL: Manual Text Submission
Title: Global Climate Summit
Word Count: 30 words | **Date:** 2026-10-24 11:56:13

VERDICT: UNCERTAIN

Calibrated Confidence: 81.0% (Threshold tau = 65.0%)
Base Disagreement (std): 0.295 (Threshold delta = 0.20)
Gating Trigger: High Disagreement

CAUTION: Short Text Length (30 words)

Input text contains only 30 words (< 50 words recommended). Stylometric distributions (type-token ratio, hedge frequency, attribution density) and readability formulas exhibit higher variance on short passages; article-level diagnostics should be interpreted with caution.

1. Multi-Model Disagreement Analysis (5 Models)

Model Name	P(FAKE)	Model Vote	Confidence
Random Forest	59.6%	FAKE	59.6%
XGBoost	60.1%	FAKE	60.1%
LightGBM	78.2%	FAKE	78.2%
Logistic Regression	0.1%	REAL	99.9%
CAGE-Fake Stack	19.0%	REAL	81.0%

Consensus Status: High Dissent / Conflicting Predictions
Base Disagreement (4-base std): 0.295 (Delta = 0.20)
5-Model Consensus Spread: 0.290
Vote Distribution: 3 FAKE vs 2 REAL
Diagnostic: Base-model disagreement (0.295) exceeds safety threshold delta (0.200).

2. News Credibility Fingerprint (Linguistic & Stylistic Scorecard)

Credibility Dimension	Score (0-100)	Journalistic Assessment
Sensationalism Discipline	77.9/100	Adequate
Source Attribution	72.1/100	Adequate
Epistemic Grounding	77.9/100	Adequate
Punctuation Discipline	80.7/100	Excellent
Readability Structure	77.9/100	Adequate
Lexical Diversity	75.0/100	Adequate

Overall Credibility Index: 76.4/100
Credibility Tier: B (Standard News Distribution)

Scores are normalized against verified newsroom style guides: high attribution, balanced syntax, and low sensationalism maximize the index.

3. Transparent Reasoning & Counterfactual Sensitivity

Diagnostic Explanation:

1. Decision & Triage Status: The CAGE-Fake pipeline reached a verdict of **UNCERTAIN** with **81.0% calibrated confidence**.
- **Gating Trigger (High Disagreement)**: Base-model disagreement (0.295) exceeded threshold delta (0.200). Although confidence was 81.0%, the base classifiers sharply diverged in their verdicts. The system abstains to ensure decisions require multi-model consensus.

Counterfactual Simulation: No qualifying sensational words, speculative hedges, or shout punctuation detected in the article text; no counterfactual perturbations were applied.
Original P(FAKE): 19.0% | **Transformed P(FAKE):** 19.0% **(Signed Shift: +0.0 percentage points)**

4. Local Signal Attributions (Meta-Ensemble Impacts)

Top FAKE-Driving Indicators	Weight / Impact	Top REAL-Driving Anchors	Weight / Impact
LightGBM [P(FAKE)]	+0.377	Logistic Regression [P(FAKE)]	-2.349
Random Forest [P(FAKE)]	+0.229	Base Disagreement (std)	-0.103
XGBoost [P(FAKE)]	+0.102	-	-

5. Empirical Benchmark Model Comparison (Held-out Test Split)

Model	Accuracy	Macro-F1	ECE	Coverage	Selective Acc
Random Forest (Uncalibrated)	0.8557	0.8552	0.1488	100.0%	0.8557
XGBoost (Uncalibrated)	0.8705	0.8703	0.0371	100.0%	0.8705
LightGBM (Uncalibrated)	0.8787	0.8785	0.0401	100.0%	0.8787
Logistic Regression (Uncalibrated)	0.8951	0.8951	0.0372	100.0%	0.8951
Uncalibrated Stack	0.8984	0.8983	0.0372	100.0%	0.8984
Calibrated Stack (Full Coverage)	0.8967	0.8967	0.0276	100.0%	0.8967
CAGE-Fake (Calibrated + Gated)	**0.8967**	**0.8967**	**0.0276**	**84.6%**	**0.9496**

6. Research Calibration & Disagreement Visualizations







